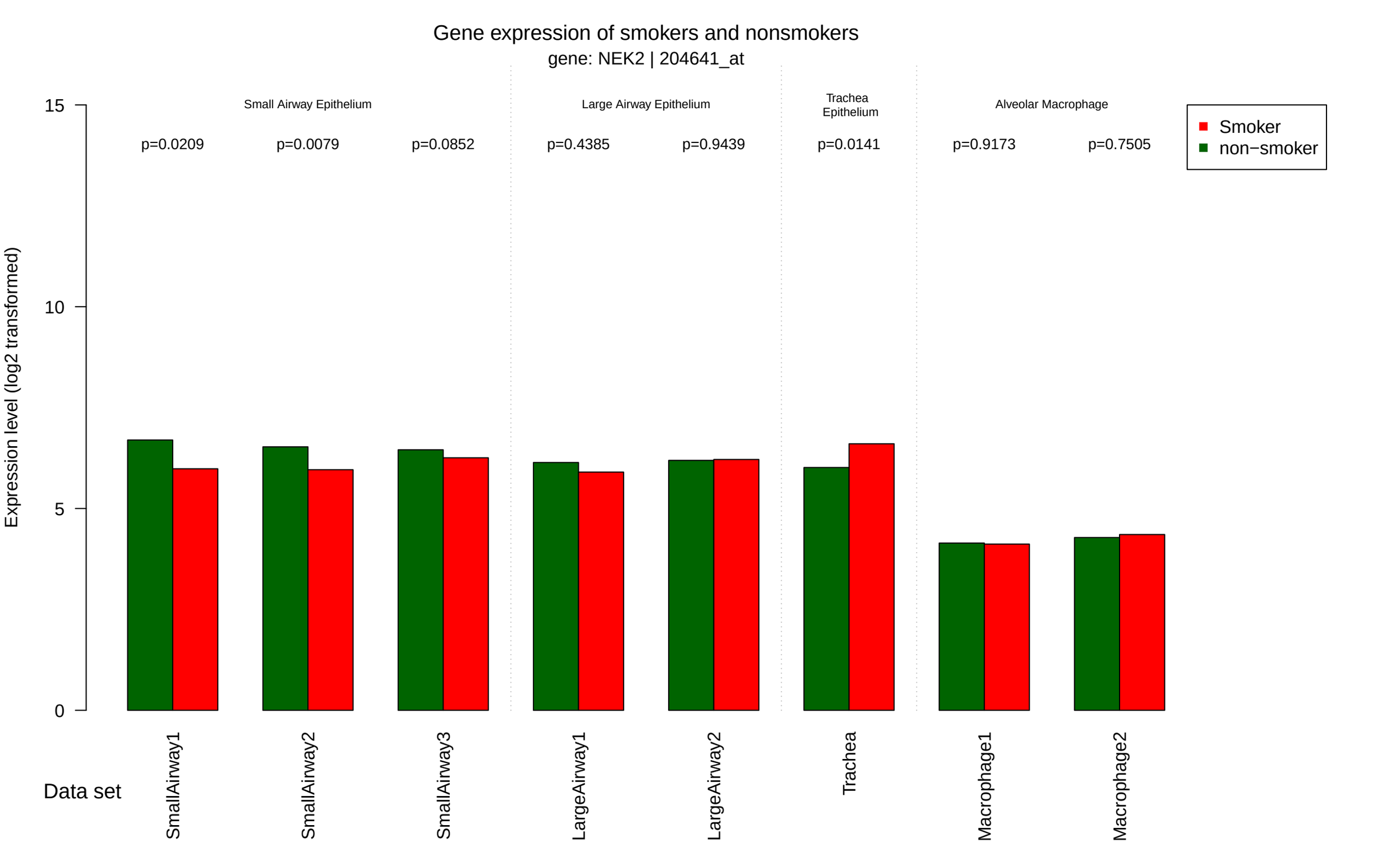
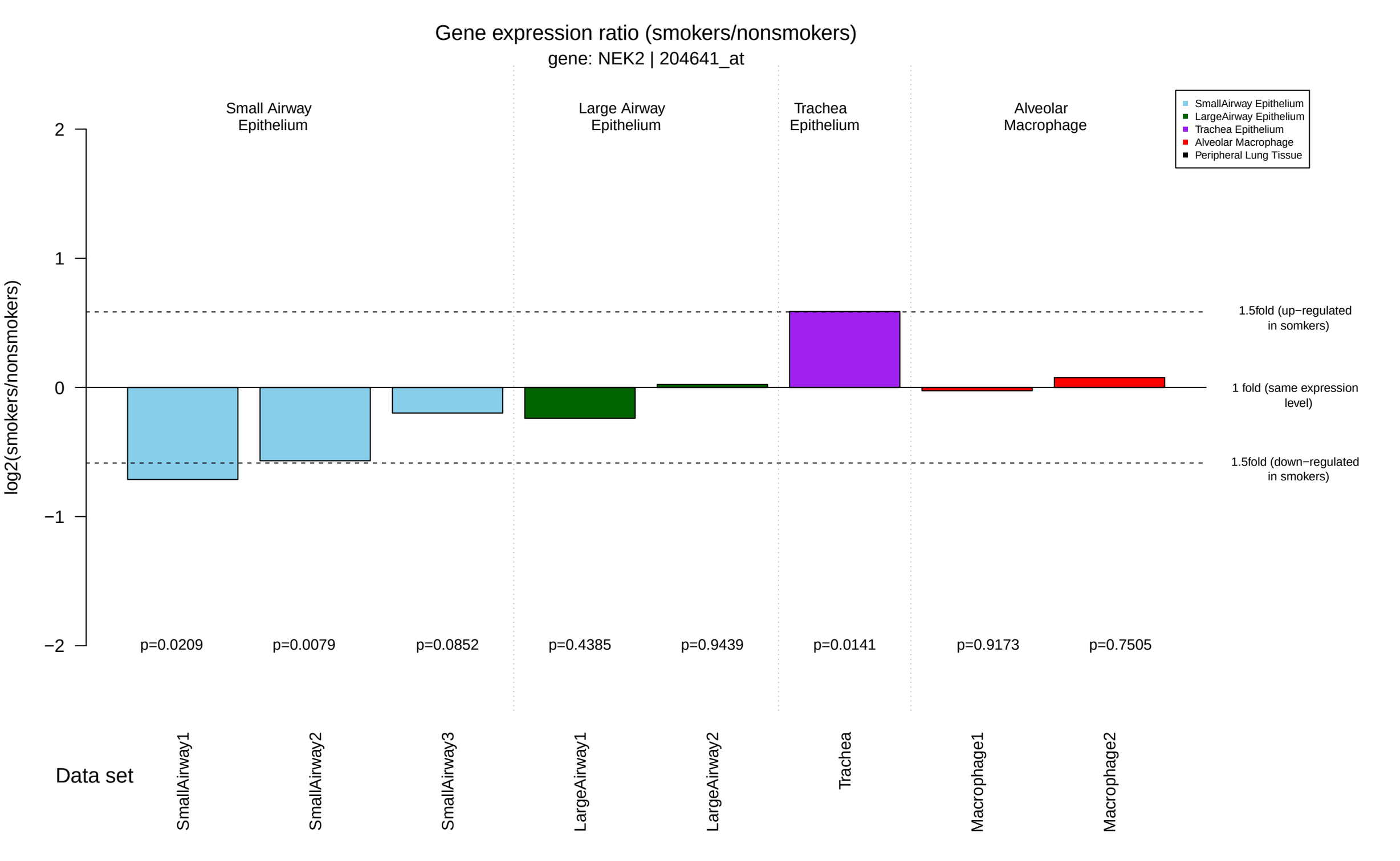
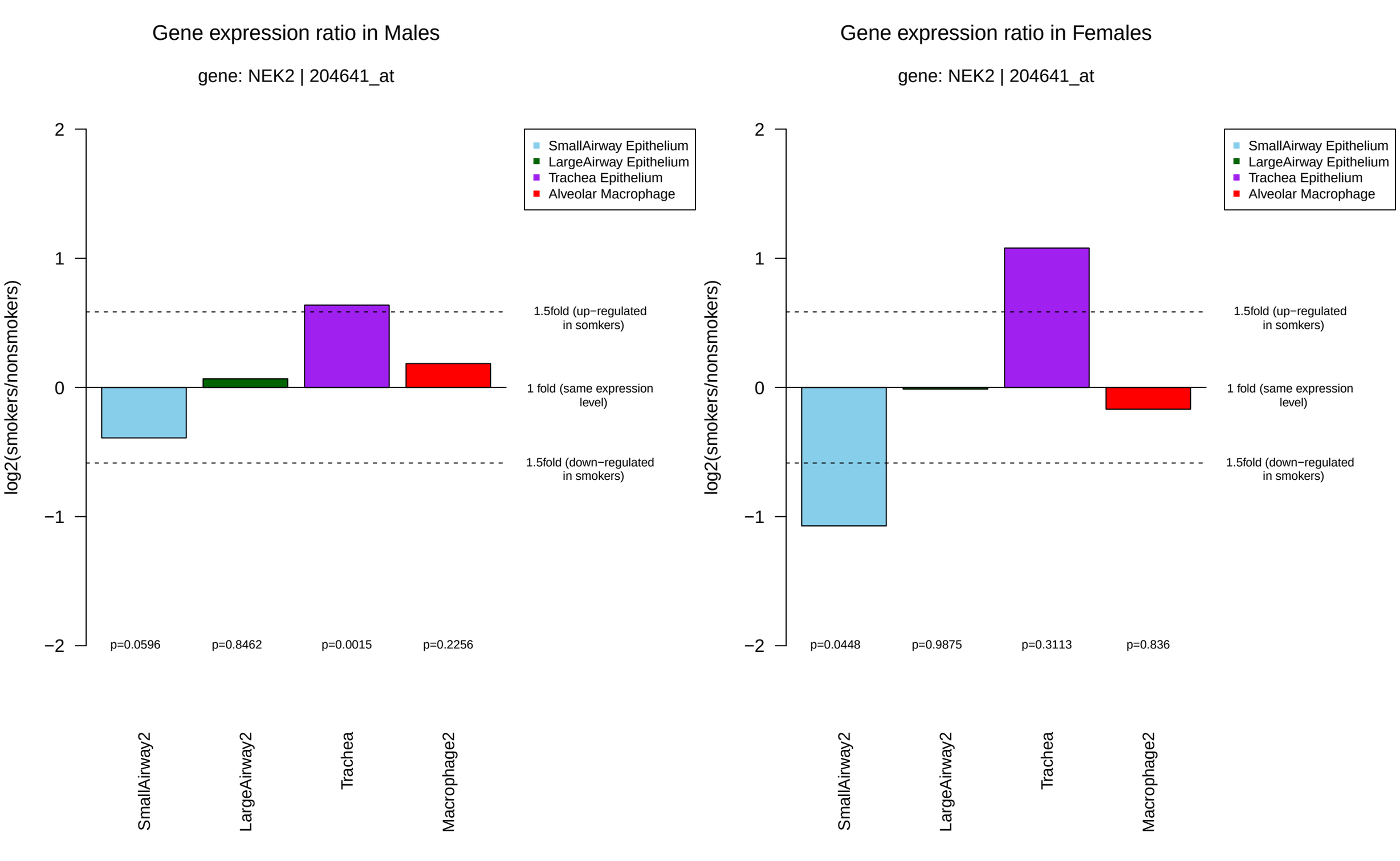
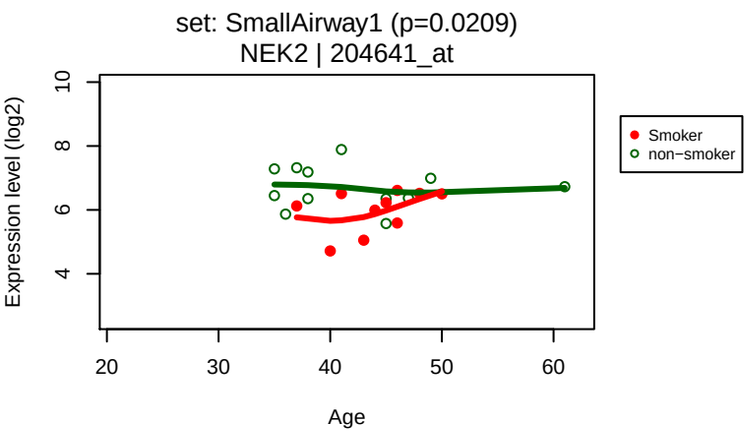
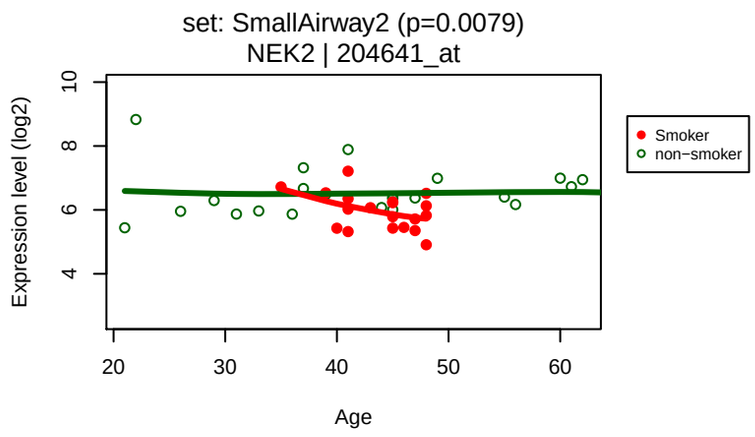
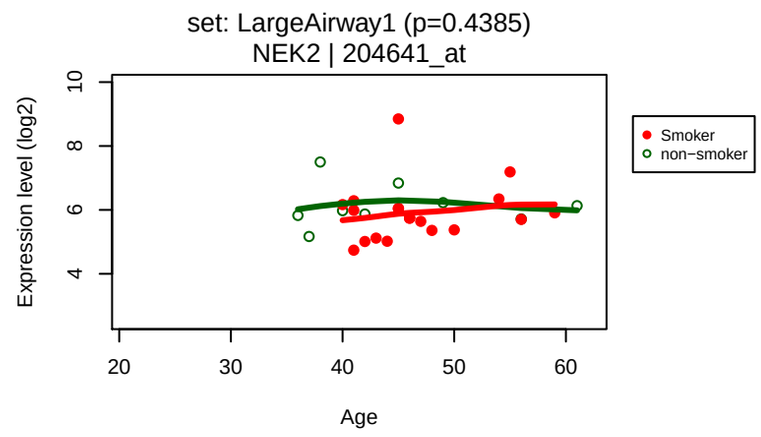

Gene expression of smokers and nonsmokers
gene: NEK2 | 204641_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0209
p=0.0079
p=0.0852
p=0.4385
p=0.9439
p=0.0141
p=0.9173
p=0.7505
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

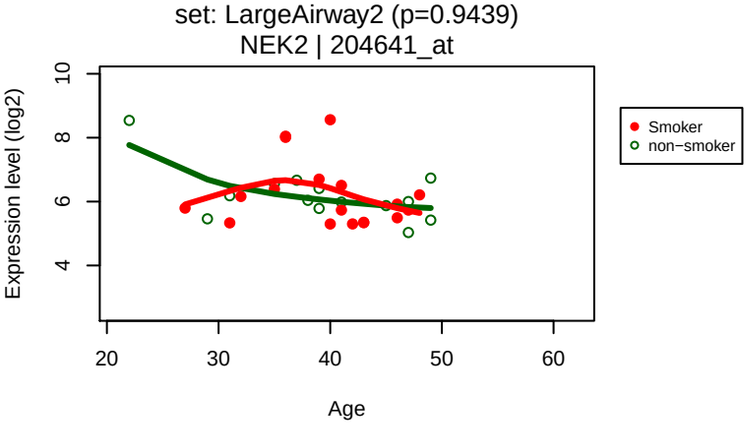

Gene expression ratio (smokers/nonsmokers)
gene: NEK2 | 204641_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0209
p=0.0079
p=0.0852
p=0.4385
p=0.9439
p=0.0141
p=0.9173
p=0.7505
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

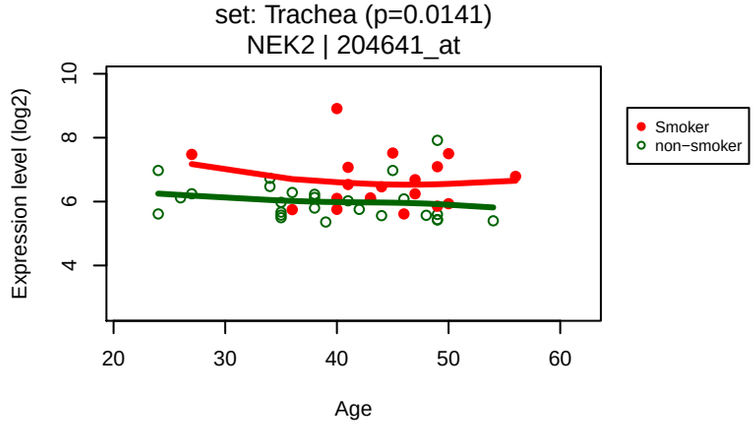

Gene expression ratio in Males
gene: NEK2 | 204641_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0596
p=0.8462
p=0.0015
p=0.2256
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NEK2 | 204641_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0448
p=0.9875
p=0.3113
p=0.836
SmallAirway2
LargeAirway2
Trachea
Macrophage2

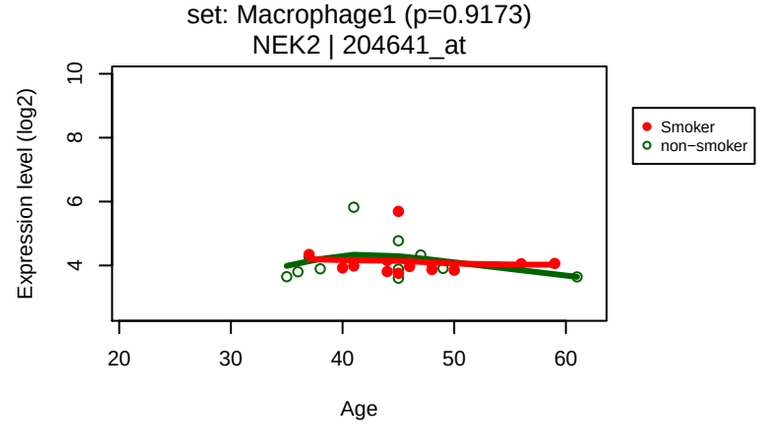

set: SmallAirway1 (p=0.0209)
NEK2 | 204641_at
Expression level (log2)
Age
Smoker
non−smoker

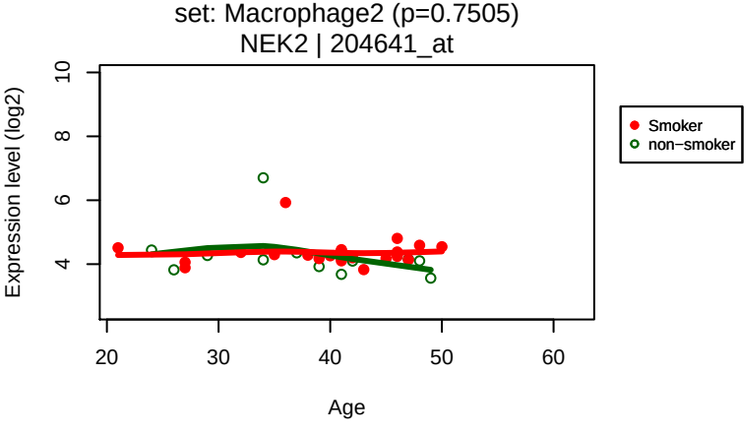

set: SmallAirway2 (p=0.0079)
NEK2 | 204641_at
Expression level (log2)
Age
Smoker
non−smoker

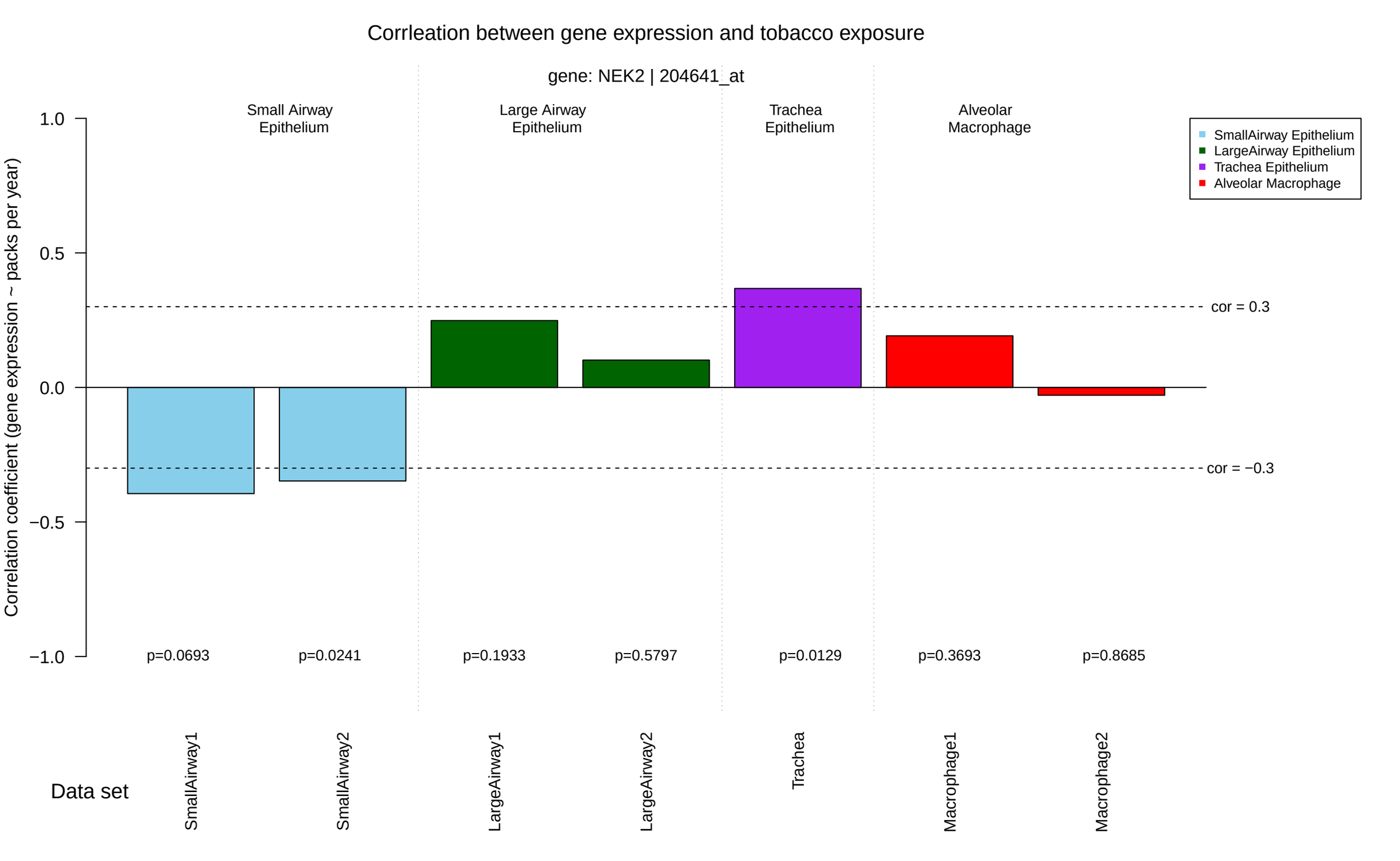

set: LargeAirway1 (p=0.4385)
NEK2 | 204641_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9439)
NEK2 | 204641_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0141)
NEK2 | 204641_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9173)
NEK2 | 204641_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7505)
NEK2 | 204641_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NEK2 | 204641_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0693
p=0.0241
p=0.1933
p=0.5797
p=0.0129
p=0.3693
p=0.8685
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2